

GRD theory, part 2 (draft for supervisor review): the gate is rate-optimal

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Draft of 7 October 2026, rechecked: detection constant corrected to 0.67

Purpose and result

Theorem 7 and Corollary 8 of the manuscript are upper bounds: the gate certifies a shifted environment once its population statistic γ exceeds about $\sqrt{d/n_e}$ (up to logarithmic factors), and the certified direction is then recovered with angle error about $\sqrt{d/n_e}/\gamma$. A reviewer’s natural question is whether the gate is too conservative. This note answers it with matching lower bounds:

- **Detection (Theorem LB1).** If $\gamma \leq \frac{1}{2}\sqrt{d/n_e}$, no test, even one that knows the control covariance exactly, can tell a shifted environment from a null one: its two error probabilities sum to at least 0.67.
- **Recovery (Theorem LB2).** No estimator recovers the unmixing direction with angle error below $0.05\sqrt{(d-2)/n_e}/\gamma$ with probability above one half.

Together with Theorem 7 and Corollary 8 (Corollary LB1), the gate and the recovery step are rate-optimal up to logarithmic factors: the gate abstains, up to constants and logarithms, only where certification and recovery are impossible for every method. This turns abstention from a conservative heuristic into the correct decision, and it is the strongest theoretical statement the paper can make within its model.

1 Novelty sweep (abbreviated)

Five search angles on 7 October 2026: lower bounds for interventional CRL; sample complexity of interventional CRL; testing or certifying identifiability conditions with abstention; minimax detection of covariance or precision changes; and rank-one spike detection.

Work	What it proves	Relation
Acartürk et al. [1]	Sample-size upper bounds for interventional CRL; identifiability assumed	No lower bound, no testing
Lee et al. [4]	Finite-sample upper bounds with few environments; cites only the counting condition $K = \Omega(d)$ as a lower bound	No statistical lower bound, no testing
Berthet and Rigollet [2]; Cai et al. [3]	Minimax detection and estimation of spiked covariances	The technique we adapt; no CRL, no admissibility
Vu and Lei [6]	Minimax rates for principal subspace estimation	Recovery lower-bound technique
Moen [5]	Minimax rates for covariance changepoint testing	Related detection problem; no CRL

Calibrated conclusion. We found no prior result that gives matching upper and lower bounds for a pre-recovery admissibility test in interventional CRL. The lower-bound *techniques* are standard and must be cited as such; the new contribution is the statement that the gate’s certification threshold and the recovery error are minimax-optimal within the paper’s model. A full twelve-angle sweep with the `novelty-check` package should be run before submission.

2 A hard subfamily inside the paper’s model

Lemma LB1 (Embedding). *For every unit vector $v \in \mathbb{R}^d$ and every $\gamma \in (0, 1]$ there is an instance of the model of Section 2.1 of the manuscript (latent noise variances in $[1/2, 2]$, invertible R , no observation noise) whose control covariance is $\Sigma_0 = I$ and in which a parentless hard intervention produces $\Omega_e = I + \gamma vv^\top$, so $\Delta_e = \gamma vv^\top$ and $\tau_e = \gamma$. Every such instance satisfies (A2) with $K = 2$.*

Proof. Take the empty latent graph ($B = 0$), all latent noise variances equal to one, an orthogonal Q whose first column is v , and $R = Q$. Then $\Sigma_0 = QQ^\top = I$ and the unmixing row of node 1 is $w_1 = R^{-\top} e_1 = v$. A hard intervention on node 1 with $\lambda'_1 = 1/(1 + \gamma)$ gives $\Theta_e - \Theta_0 = \gamma e_1 e_1^\top$, hence $\Delta_e = \gamma vv^\top$. The spectra of Σ_0 and Σ_e lie in $[1/2, 1]$. $\square$

Because R is unknown, the direction v is unknown: a lower bound may average over v . All results below hold even if the procedure is told $\Sigma_0 = I$, so they hold a fortiori for procedures that estimate Σ_0 from controls. Write $\theta = \gamma/(1 + \gamma)$, so that $\Sigma_e = I - \theta vv^\top$ and the intervention lowers the variance along v from 1 to $1 - \theta$.

3 Detection lower bound

Theorem LB1 (No test below $\sqrt{d/n}$). *Let P_0 be the law of n i.i.d. rows from $N(0, I)$ and P_v the law under $\Sigma_e = I - \theta vv^\top$. If $\gamma \leq 1$ and $\gamma \leq \frac{1}{2}\sqrt{d/n}$, then every test ϕ satisfies*

$$P_0(\phi = 1) + \sup_{\|v\|=1} P_v(\phi = 0) \geq 1 - \frac{1}{2}\sqrt{\sqrt{2} - 1} > 0.67.$$

More generally the left side is at least $1 - \frac{1}{2}\{(1 - 2n\theta^2/d)^{-1/2} - 1\}^{1/2}$ whenever $2n\theta^2 < d$.

Proof. Let $\bar{P} = \mathbb{E}_v P_v$ with v uniform on the sphere. For any test, $P_0(\phi = 1) + \sup_v P_v(\phi = 0) \geq P_0(\phi = 1) + \bar{P}(\phi = 0) \geq 1 - \text{TV}(P_0, \bar{P}) \geq 1 - \frac{1}{2}\sqrt{\chi^2(\bar{P}||P_0)}$.

The χ^2 divergence. With $L_v = dP_v/dP_0$ for one row, $L_v(x) = \sqrt{1 + \gamma} \exp\{-\gamma(v^\top x)^2/2\}$, and a Gaussian integral gives $\mathbb{E}_0[L_v L_{v'}] = (1 + \gamma) \det(I + \gamma vv^\top + \gamma v'v'^\top)^{-1/2} = (1 - \theta^2 t)^{-1/2}$ with $t = \langle v, v' \rangle^2$. For n i.i.d. rows, $\chi^2(\bar{P}||P_0) = \mathbb{E}_{v, v'}(1 - \theta^2 t)^{-n/2} - 1$.

Bounding the expectation. Since $\gamma \leq 1$, $\theta \leq \frac{1}{2}$ and $\theta^2 t \leq \frac{1}{4}$, and $-\log(1 - x) \leq 2x$ on $[0, \frac{1}{2}]$ gives $(1 - \theta^2 t)^{-n/2} \leq \exp(n\theta^2 t)$. For independent uniform v, v' , $t \sim \text{Beta}(\frac{1}{2}, \frac{d-1}{2})$, whose moment generating function is the confluent hypergeometric function ${}_1F_1(\frac{1}{2}; \frac{d}{2}; s) = \sum_k (\frac{1}{2})_k s^k / \{(\frac{d}{2})_k k!\}$. Since $(\frac{d}{2})_k \geq (\frac{d}{2})^k$, it is bounded termwise by $\sum_k (\frac{1}{2})_k (2s/d)^k / k! = (1 - 2s/d)^{-1/2}$ for $s < d/2$. With $s = n\theta^2$, $\chi^2 \leq (1 - 2n\theta^2/d)^{-1/2} - 1$. Under the stated conditions $\theta \leq \gamma \leq \frac{1}{2}\sqrt{d/n}$, so $2n\theta^2/d \leq \frac{1}{2}$ and $\chi^2 \leq \sqrt{2} - 1$. $\square$

4 Recovery lower bound

Theorem LB2 (No estimator below $\sqrt{d/n}/\gamma$). *Let $d \geq 6$ and $\gamma \in (0, 1]$. For any estimator $\hat{v}$ computed from n rows,*

$$\sup_{\|v\|=1} P_v \left(\sin \angle(\hat{v}, v) \geq 0.05 \min \left\{ 1, \frac{1}{\gamma} \sqrt{\frac{d-2}{n}} \right\} \right) \geq \frac{1}{2}.$$

Proof. Divergence. For unit v, v' , $\Sigma_v = I - \theta vv^\top$ and $\Sigma_{v'}^{-1} = I + \gamma v'v'^\top$ have equal determinants, and $\text{tr}(\Sigma_{v'}^{-1} \Sigma_v) = d - \theta + \gamma - \theta\gamma \langle v, v' \rangle^2$. Since $\gamma - \theta = \theta\gamma$, the Kullback–Leibler divergence of n rows is $\text{KL}(P_v \| P_{v'}) = \frac{n}{2} \theta \gamma \sin^2 \angle(v, v') \leq \frac{n}{2} \gamma^2 \sin^2 \angle(v, v')$.

Packing. The unit sphere of $e_1^\perp$ contains $M \geq 2^{d-2}$ points u_j with $\|u_j - u_k\| \geq \frac{1}{2}$ (a volume argument on the sphere S^{d-2}). For $\varepsilon \in (0, 1/\sqrt{2}]$ set $v_j = \sqrt{1 - \varepsilon^2} e_1 + \varepsilon u_j$. Then $\langle v_j, v_k \rangle = 1 - \varepsilon^2 \|u_j - u_k\|^2/2 \in [0, 1 - \varepsilon^2/8]$, so $\sin^2 \angle(v_j, v_k) \geq 1 - \langle v_j, v_k \rangle \geq \varepsilon^2/8$, and $\sin^2 \angle(v_j, v_k) \leq \|v_j - v_k\|^2 \leq 4\varepsilon^2$.

Fano. The pairwise divergences are at most $2n\gamma^2\varepsilon^2$. Choose $\varepsilon^2 = \min\{\frac{1}{2}, (d-2) \log 2/(8n\gamma^2)\}$. Then $\max \text{KL} \leq \frac{1}{4} \log M$, and Fano’s inequality gives an identification error of at least $1 - (\frac{1}{4} \log M + \log 2)/\log M \geq \frac{1}{2}$ for $d \geq 6$. Because $\sin \angle$ is a metric on lines and the v_j are $\varepsilon/\sqrt{8}$ apart, any $\hat{v}$ with $\sin \angle(\hat{v}, v_j) < \varepsilon/(2\sqrt{8})$ identifies j ; so with probability at least one half the error is at least $\varepsilon/(2\sqrt{8}) \geq 0.05 \min\{1, \gamma^{-1} \sqrt{(d-2)/n}\}$. $\square$

5 Rate optimality

Corollary LB1 (The gate is rate-optimal up to logarithms). *Within the model class of Lemma LB1 ($K = 2$), and with $L, \bar{\varepsilon}$ as in Theorem 7 of the manuscript:*

- (a) *the gate certifies every shifted environment with probability at least $1 - \delta$ once $(B_{\text{boot}} + 1)kq \geq m$ and*

$$\gamma \geq C \sqrt{(d+L)L/n_{\min}} \quad (\text{Theorem 7}),$$

while no test can do so reliably when $\gamma \leq \frac{1}{2} \sqrt{d/n_e}$ (Theorem LB1);

- (b) *the certified direction is recovered with angle error*

$$\sin \angle \leq 4\bar{\varepsilon}/\gamma \leq C \sqrt{(d+L)L/n_{\min}} / \gamma \quad (\text{Corollary 8}),$$

while no estimator achieves $0.05 \sqrt{(d-2)/n_e} / \gamma$ uniformly (Theorem LB2).

The upper and lower thresholds differ by a factor of order L , logarithmic in the number of environments, null draws, samples, and $1/\delta$.

What this does not say. Optimality is in the rate, not the constant; at finite samples the constants matter, and the positive control of the manuscript (certification only from signal-to-background ratio 4) is consistent with a constant of a few units. The lower bounds are worst case over directions within the hard-intervention subclass. They say nothing about attribution: a measurement change with $\tau_e \geq \gamma$ is as detectable as a mechanism change (Proposition 5 of the manuscript).

6 Numerical check

Setting of Lemma [LB1](#), $d = 10$, $\gamma = x\sqrt{d/n_e}$ for $x \in \{0.25, \dots, 6\}$ and $n_e \in \{200, 800, 3200\}$. “Oracle” is the test $\lambda_{\max}(\hat{\Omega}_e - I)$ that knows $\Sigma_0 = I$, calibrated on 2,000 null simulations (60 replicates); “gate” is the committed GRD gate with $n_0 = 4n_e$ estimated controls, disjoint null, $B_{\text{boot}} = 199$ (30 replicates); “sin” is the gate’s median recovery error.

$x = \gamma\sqrt{n_e/d}$	0.25	0.5	1	2	3	4	6
Oracle power, $n_e = 200$	0.07	0.12	0.15	0.43	0.72	0.95	1.00
Oracle power, $n_e = 800$	0.07	0.07	0.23	0.60	0.97	1.00	1.00
Oracle power, $n_e = 3,200$	0.02	0.05	0.17	0.75	0.98	1.00	1.00
Gate power, $n_e = 200$	0.10	0.07	0.10	0.40	0.60	1.00	1.00
Gate power, $n_e = 800$	0.03	0.13	0.13	0.53	0.93	0.97	1.00
Gate power, $n_e = 3,200$	0.00	0.07	0.07	0.67	1.00	1.00	1.00
Gate sin, $n_e = 200$	0.97	0.95	0.89	0.58	0.44	0.35	0.25
Gate sin, $n_e = 800$	0.96	0.93	0.80	0.50	0.35	0.32	0.21
Gate sin, $n_e = 3,200$	0.93	0.95	0.82	0.55	0.37	0.27	0.19

The curves collapse on x , as the theorems predict: at $x \leq 0.5$ neither the oracle nor the gate has power above 0.13, within the bound of Theorem [LB1](#), the transition sits between $x = 1$ and $x = 3$ for every n_e , the gate’s power is close to the oracle’s, and $x \cdot \text{sin}$ is roughly constant once the environment is detectable.

7 Cost to the paper

Two theorems and a corollary (about one page in Section 3.5, proofs in Appendix D), one sentence in the abstract and contributions, and the simulation above as a small figure or table. No new real-data runs. The gate code is unchanged.

References

- [1] Emre Acartürk, Burak Varıcı, Karthikeyan Shanmugam, and Ali Tajer. Sample complexity of interventional causal representation learning. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- [2] Quentin Berthet and Philippe Rigollet. Optimal detection of sparse principal components in high dimension. *The Annals of Statistics*, 41(4):1780–1815, 2013.
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- [4] Inbeom Lee, Tongtong Jin, and Bryon Aragam. Beyond identifiability: Learning causal representations with few environments and finite samples. arXiv:2603.25796, 2026.
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